

Sir Syed University of Engineering & Technology
Department of Computer Science & Information Technology



CS-467- Software Project Management

Lecture No. 1

Introduction to Software Project Management

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List of Topic to be Discuss in this lecture

- What is Management?
- What is a Project?
- What is Software?
- What is Project Management?
- Differentiating Factors of Software Project
- What is Software Project Management?
- Comparing Projects and Operations
- Project Management Framework
- Factors Used to Differentiate Projects
- Types of Software Projects

What is Management?

Basically, the **management involves** the following activities:

1. **Planning-** deciding what is to be done
2. **Organizing-** making arrangements
3. **Staffing-** selecting the right people for the job
4. **Directing-** giving instructions
5. **Monitoring-** checking on progress
6. **Controlling-** taking action to remedy hold-ups
7. **Innovating-** coming up with new solutions
8. **Representing-** liaising with users, etc.

What is a Project?

A project is an activity with specific goals which takes place over a finite period of time.

Examples of projects include:

- **Effecting a change** in structure, staffing, or style of an organization.
- **Designing** a new transportation vehicle.
- **Developing** or **acquiring** a new or modified information system.
- **Constructing** a building or facility.
- **Running** a campaign for political office.
- **Implementing** a new business procedure or process.

What is Software?

Software is Computer **programs**, **procedures**, and possibly **associated documentation** and **data pertaining** to the **operation** of a computer system.

The **IEEE definition** of software, which is almost identical to the **ISO definition** (ISO, 1997, Sec. 3.11 and ISO/IEC 9000-3 Sec. 3.14), lists the following **four components** of software:

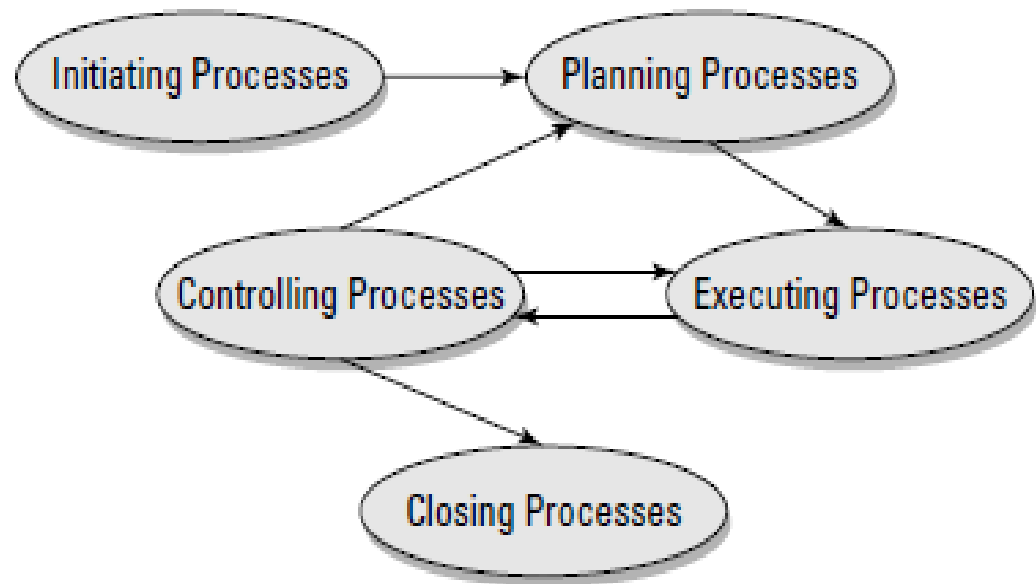
1. Computer programs (the “code”)
2. Procedures
3. Documentation
4. Data necessary for operating the software system.

What is Project Management?

Project Management is the application of knowledge, skills, tools and techniques to project activities to meet project requirements.

Project management is accomplished through the **use of the processes** such as:

1. Initiating
2. Planning
3. Executing
4. Monitor and Controlling
5. Closing



What is Project Management?

- ✓ **Initiating:** That's really where you are now. The project is in the process of getting selected, sponsored, funded, and launched.
- ✓ **Planning:** As you can see in figure above, planning is an iterative process. Planning basically determines how the project work will get accomplished.
- ✓ **Executing:** After you get a plan, your project team does the work.
- ✓ **Monitoring and Control:** Your project team does the work, but you control them.
- ✓ **Closing:** After the project work has been completed, you tie up loose ends and close out the software project.

Differentiating Factors of Software Project

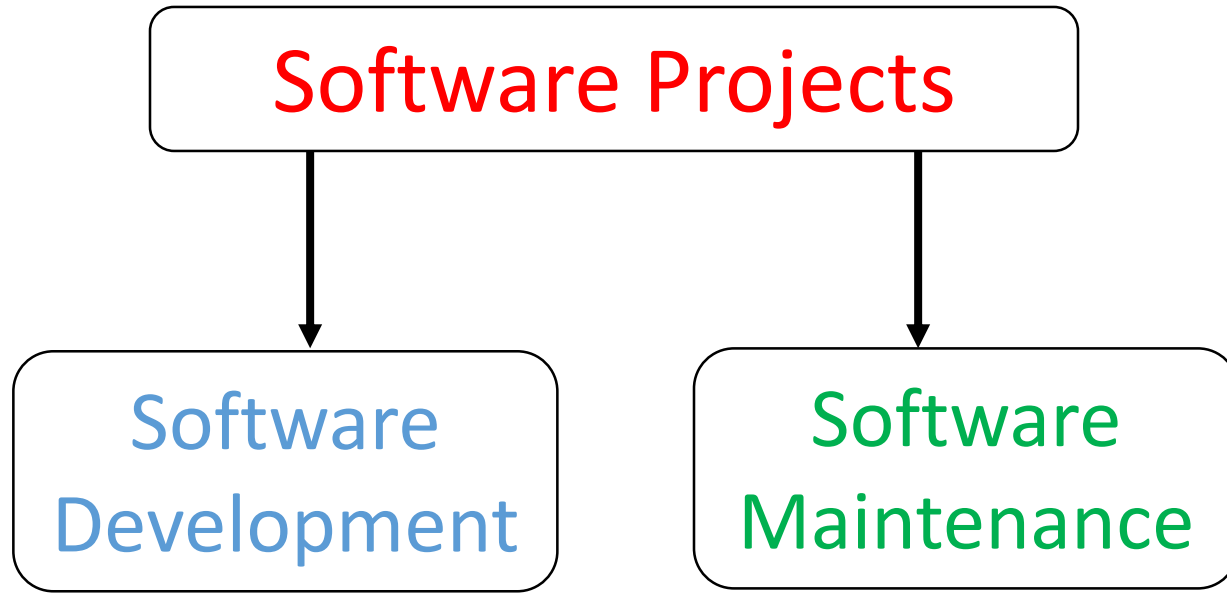
Following some of the main differentiating factors are as follows:

- ❑ Software is an **intangible** product.
- ❑ Software is a **cognitive** and **human-based** development process that requires **sharing of documents**.
- ❑ There is a **higher degree** of **uncertainty** in the project and product scope.
- ❑ **Communication** and **coordination** within **software teams** and with **project stakeholders** often **lacks clarity**.
- ❑ The **intellectual capital** of software personnel is the **primary asset** of software projects and organizations..

Differentiating Factors of Software Project

- ❑ Initial planning and estimation of software projects is challenging because these activities depend on requirements that are often imprecise or based on lacking information.
- ❑ Exhaustive testing of software is impractical because of the time and related complexity constraints.
- ❑ Software security is a large and growing challenge.
- ❑ Objective measurement and quantification of software quality is difficult.

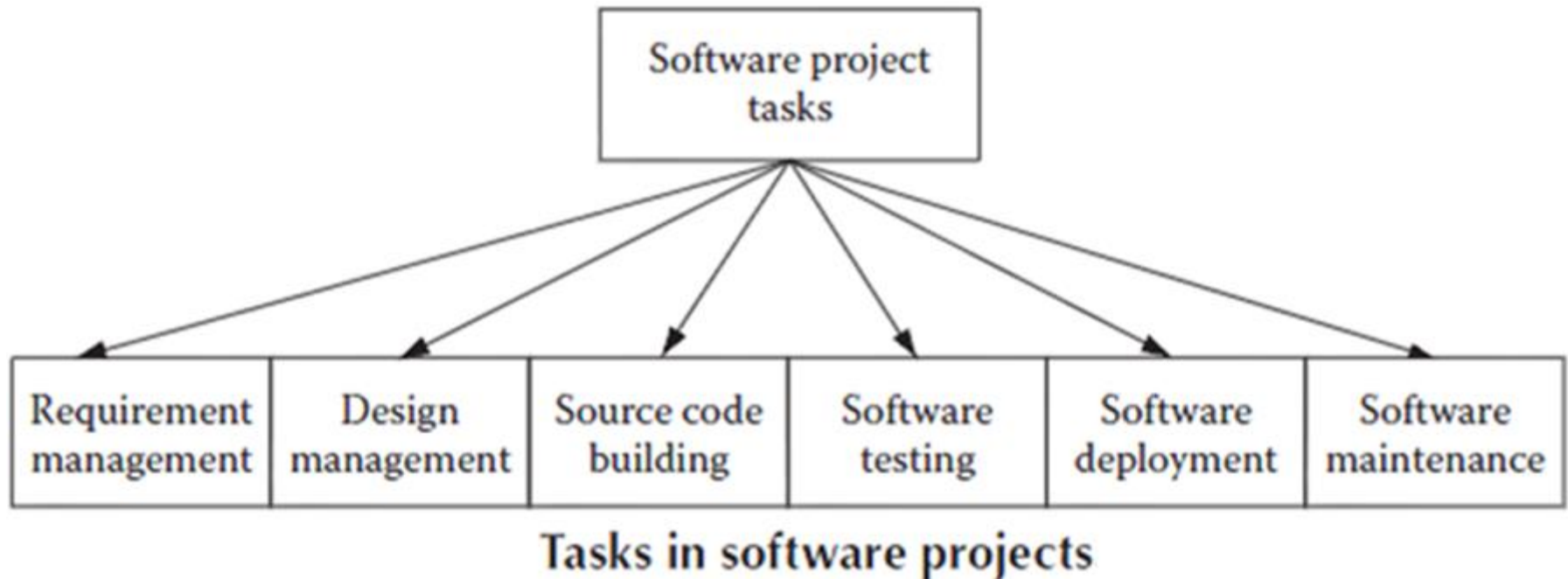
Software Development and Maintenance Project



Software Development and Maintenance Project

- ❑ In **software development project** on the other hand is making software design based on customer requirements and implementing it into source code. This **source code** is then **tested** to make sure that it is defect free so that end users can use the software system without running into many problems.
- ❑ In **software maintenance project**, an **already existing software** product is **modified** to remove software defects, **add new functionality**, port the software product on some other operating system, etc.
- ❑ Software development and software maintenance projects together are referred to as **software projects**.

Software Development and Maintenance Project



What is an Information Technology (IT) Project?

Information technology (IT) is the use of any computers, storage, networking and other physical devices, infrastructure and processes to create, process, store, secure and exchange all forms of electronic data.

Information Technology

- Software System
- Hardware System
- Telecommunication

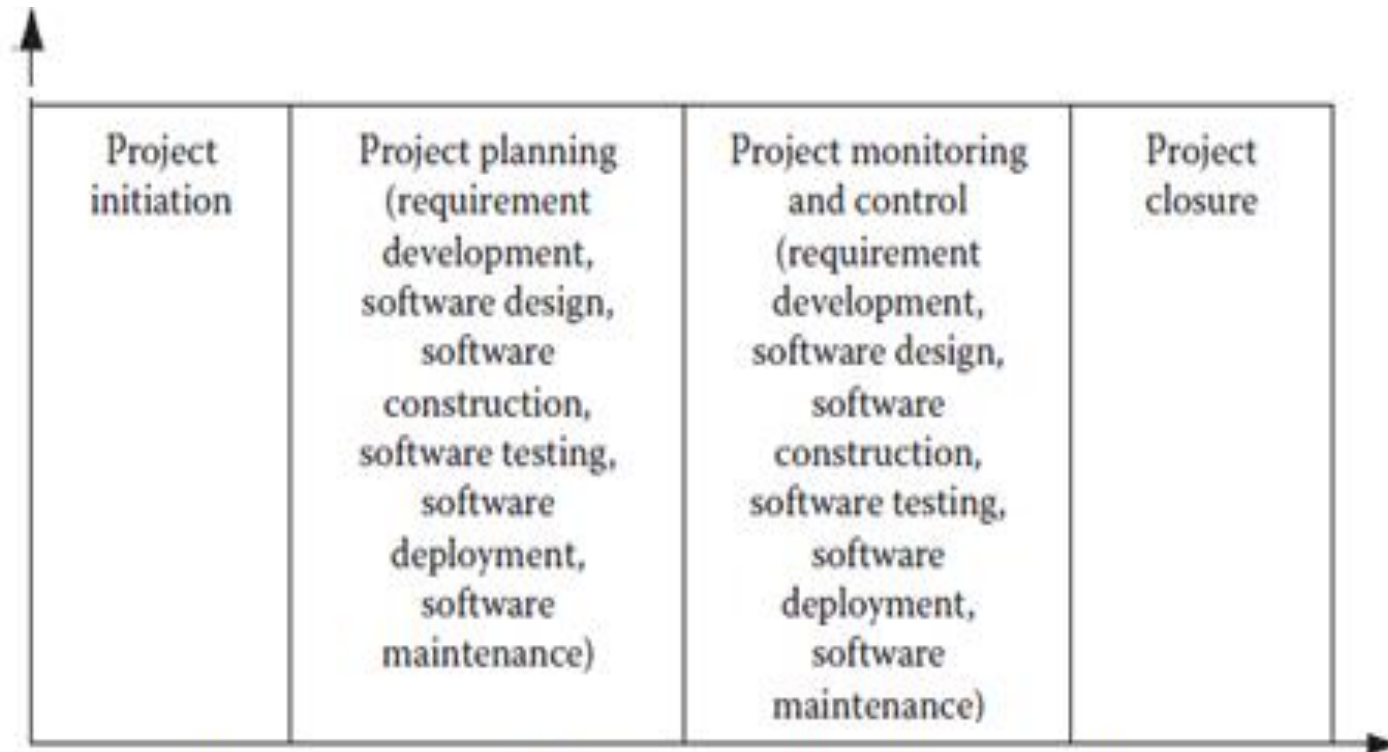
For example, a bank's ATM machine use for doing transactions may be a complete IT system when the software system is installed in the hardware system, Transactions require a communication system and is ready to be used by end users.

What is Software Project Management?

- ❑ In a nutshell, software project management can be defined as applying project management and software engineering methods to develop/maintain a software product so that the goal of developing/maintaining a software product can be achieved using minimum possible resources and money and within the minimum time possible.
- ❑ For instance, a project manager responsible for managing a civil construction project must have knowledge and experience in civil engineering. An electrical engineer managing a civil project will not be a suitable fit. Similarly, a project manager responsible for managing a software project must have knowledge and experience in software engineering.

What is Software Project Management?

Project management processes may include project initiation, project planning, project monitoring and control, and finally project closure. The software engineering processes may include requirement development, software design, software construction, software testing, and software maintenance.



Software project management processes with software engineering processes

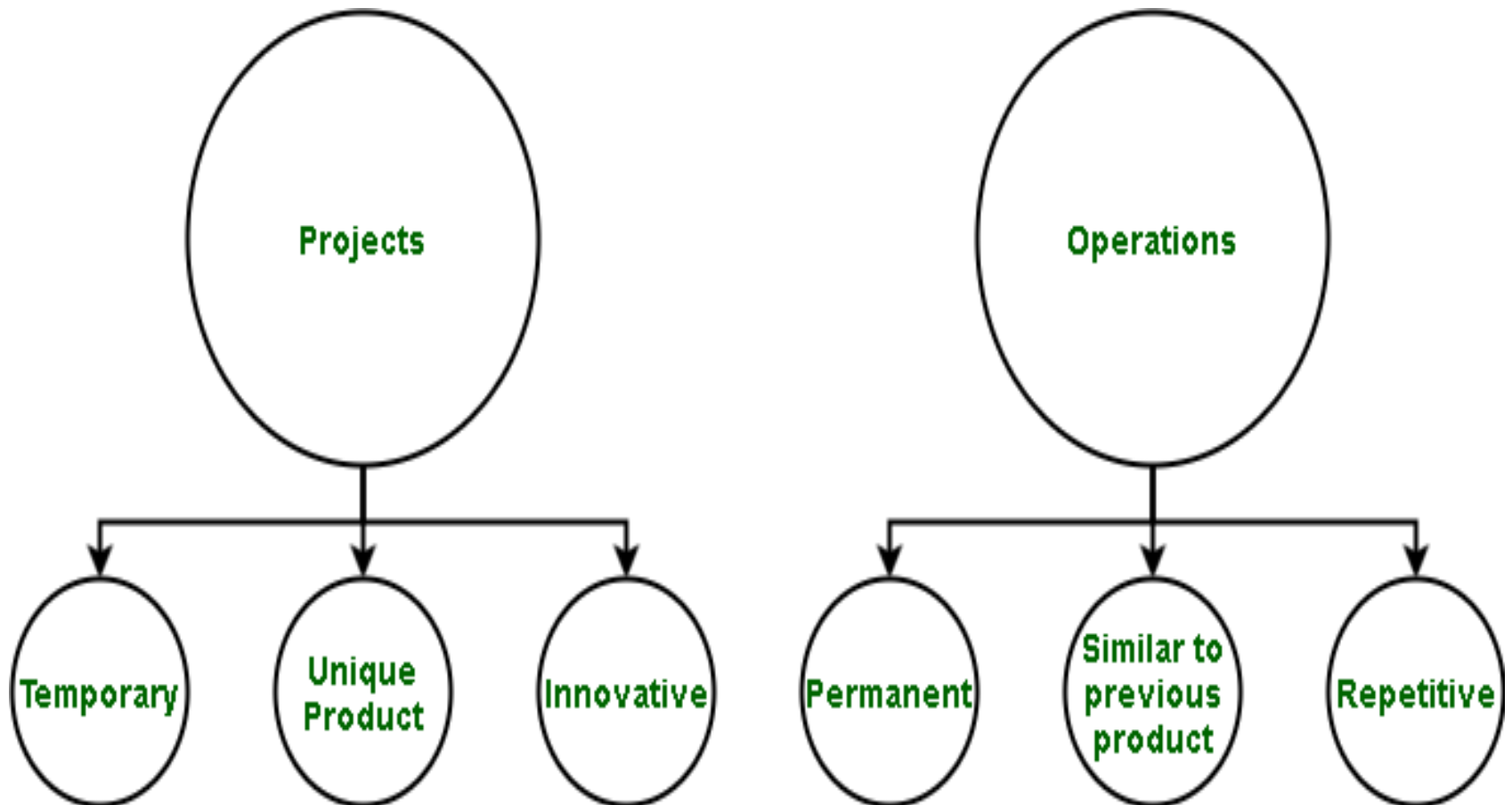
Comparing Projects and Operations

- ❑ **Operations** are the **day-to-day activities** that your organization does.
- ❑ **Project** is a **temporary endeavour** (عارضی کوشش) completed to create a unique product, service and result.

Example

- ❑ Developing a **new product** is a **project** — but the **continued production of the product** is an **operation**. Operational activities may repeat as needed, daily, monthly, or annually.

Comparing Projects and Operations



What Makes Software Project Management So Special

What is special about project management, however, is the nature of the work. Just as the particulars of designing a new warehouse, building a house, or creating a prototype for a remote controlled airplane are unique, so is the creation of software:

- ❑ Software development is **weird**(عجیب) and requires a **specialized skill** set to do it well.
- ❑ Software creation is **tough**.
- ❑ Software development can be **boring, routine, and mind numbing**(دماغ بے حسی).
- ❑ Software creation can create **challenges** within the development of the code.

Project Management Framework

**Stakeholder
needs and
expectations**



Process groups	Knowledge areas	Tools and techniques
1. Initiating	1. Integration	
2. Planning	2. Scope	
3. Executing	3. Schedule	
4. Monitoring and controlling	4. Cost	
5. Closing	5. Quality	
	6. Resource	
	7. Communication	
	8. Risk	
	9. Procurement	
	10. Stakeholder	

Project Management Framework

PMI has developed a Project Management Framework which provides a basic structure and Knowledge Areas to describe project management body of knowledge and practices in terms of their component processes. These processes have been organized into **10 knowledge areas**:

- **Project Integration management:** the processes and activities needed to identify, define, combine, unify, and coordinate the various processes and project management activities within the project management process groups.
- **Project Scope Management:** the processes required to ensure that the project includes all the work required, and only the work required, to complete the project successfully.
- **Project Time Management:** the processes required to manage the timely completion of the project.

Project Management Framework

- **Project Cost Management:** the processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling costs so that the project can be completed within the approved budget.
- **Project Quality Management:** the processes and activities of the performing organization that determine quality policies, objectives, and responsibilities so that the project will satisfy the needs for which it was undertaken.
- **Project Human Resource Management:** the processes that organize, manage, and lead the project team.
- **Project Communications Management:** the processes that are required to ensure timely and appropriate planning, collection, creation, distribution, storage, retrieval, management, control, monitoring, and the ultimate disposition of project information.

Project Management Framework

- **Project Risk Management:** the processes of conducting risk management planning, identification, analysis, response planning, and controlling risk on a project.
- **Project Procurement Management:** the processes necessary to purchase or acquire products, services, or results needed from outside the project team. Processes in this area include Procurement Planning, Solicitation Planning, Solicitation, Source Selection, Contract Administration, and Contract Closeout.
- **Project Stakeholder Management:** the processes required to identify all people or organizations impacted by the project, analyzing stakeholder expectations and impact on the project, and developing appropriate management strategies for effectively engaging stakeholders in project decisions and execution.

Factors Used to Differentiate Projects

Organization Level Factors:

- **Business domain:** For what domain is the software (based) product developed?
- **Number of instances:** How many instances of the software (based) product will be deployed?
- **Maturity of organization:** How mature are the processes of the software developing organization?
- **Level of innovation:** How innovative is the organization?
- **Culture:** In which culture are the projects developed?

Factors Used to Differentiate Projects

Project Level Factors:

- **Size:** How big is the system under development?
- **Stable architecture:** Is a stable architecture in place?
- **Business model:** Under which business model is the software (–based) product developed?
- **Team distribution:** How many teams are working in the project and in which configuration?
- **Rate of change:** How stable is your business environment and how many risks and uncertainties are you facing?
- **Age of system:** Greenfield (from scratch) vs. brownfield (evolving) software system development?
- **Criticality:** How many people's safety will be threatened if the system fails?
- **Governance:** Who manages the project managers? How much governance is applied to the project?

Types of Software Projects

From the project management point of view, we can distinguish the following types of projects.

Bespoke Software: Software created to meet an individual customer's needs. Bespoke software is limited to the specific requirements of a single customer with the cost borne by the individual customer. Some examples of bespoke systems include a luggage tracking software, a compiler for a specific hardware platform, and a system to monitor a fleet of trucks. For bespoke systems, the specification of the application to develop (more in general of the project goals) will be driven and have to be agreed with the customer.

Types of Software Projects

Off-the-shelf Software: are standardized software applications that are mass-produced, available to the general public, and fit for immediate use. They are designed for a broad range of customers, offering a comprehensive set of features to streamline operations. An example of a customized off-the-shelf application is an enterprise resource planning (ERP) system.

Installation and training services: are services related to the installation of specific software systems (also in the open source domain) and/or training in the use of specific technologies or systems.

Types of Software Projects

Process and Systems Reengineering Services: are projects related to improving the efficiency of an organization, by changing the way in which they conduct their operational work.

Consider the example of a multinational company revising its customer helpdesk to improve quality and responsiveness. This project requires to understand how the organization currently works, what are the bottlenecks, and thus the possible interventions. These could include modifications to the current practices, training, and perhaps the introduction of a customer relationship management system to support the new processes. System integration services are projects and services related to automating the information flow among the different and independent systems used by an organization. The goals are to improve the efficiency of work and to reduce data duplication and errors.